

Applied Sequence Analysis

Summer term 2026

Assignment 4 A/B

Due A - 2026-05-22 12:00 PM and

B - 2026-06-03 12:00 PM

In this assignments you will plan and implement a workflow to analyse some sequenced virus genomes (Human Adenovirus).

We want:

- the assembled genomes
- a phylogenetic tree of the assembled genomes
- regions of high and/or low sequence variability

The deliverables are:

- fasta file with the assembled genome for each sequenced strain.
- phylogenetic tree in Newick format and visualised in pdf or png format.
- text file containing the average sequence variability for each window of length x (x should be a parameter of your workflow)
- plot showing the variability along the sequenced genome (x-axis: window start position, y-axis: avg. variability for window)

Your workflow of assignment 2+3 is already a good starting point, covering some necessary pre-processing and quality control steps. However, in order to expand your horizons find and use some alternative to ~~Trimmomatic~~ and (in case you need a mapper) ~~Bowtie2~~.

A - Workflow design *¹

Outline a workflow for the requested analysis, listing the tools you are planning to use for each step. Follow scientific standards, use proper citations and aim for an overall appealing appearance (not just a bullet point list or table)!

There are some questions that must be addressed:

- reference based assembly or de-novo assembly?
- how to measure sequence variability of a single sequence position?

After submission we will discuss it in-person and you can then start implementing the workflow. Doing this any time before the due date is also fine and encouraged!

B - Workflow implementation *²

- Implement your workflow according to our already established guidelines (folder structure, config file, sample sheet).
- Take care of good portability. You are now free to use all existing Snakemake wrappers. If no wrapper exists use Conda.

¹Submission: Upload a short report describing your planned workflow via KVV until 2026-05-22 12:00 PM (Assignment 4A)

²Submission: Upload your workflow via KVV until 2026-06-03 12 PM (Assignment 4B)